

Handwriting form: start here

Open this PDF in xournal-qt and write on it with the pen. Copy each printed text by hand into the box that belongs to it, once, in your normal handwriting.

This form collects your handwriting with known texts: every box expects exactly the text printed above it (or beside it). Written in xournal-qt, it becomes training data for a handwriting model of your own hand, and a test of the whole handwriting search: lines, sizes, angles, words, drawings and formulas.

- Write in your own normal hand. Copy the printed text; do not trace it.
- One box, one text: write all of it inside its box, starting at the left end, or where the arrow starts.
- The height of a box shows the size to write at; the faint line is the baseline. A box without a line is meant to be written without one.
- An arrow shows the writing direction. Turn the page or the tablet as you like.
- A mistake: strike it out and write on, or erase it. Skip a box rather than guess.
- Grey numbers (like 3.12) are box ids: leave them alone. The core form takes about 30 minutes.

If you share this form

Sharing is optional and decided later. If you might give this filled form to a common handwriting dataset, tick the licence for your handwriting and choose a writer id (any name or code).

- ☐ CC0 1.0: no rights reserved (public domain dedication) 1.1
- ☐ CC BY 4.0: free to use with credit to my writer id 1.2
- ☐ Private: for my own models only, not for sharing 1.3

Writer id 1.4	Writing hand (left or right) 1.5	Device and pen 1.6

Warm-up

Warm up: this is my handwriting on a tablet. 1.7

Notes taken on a Monday morning, with a pen. 1.8

Small steps, clear letters, steady lines. 1.9

A. Characters

Copy each group as printed, with the spaces between the characters. The typewriter font only shows clearly which character is meant; write in your normal hand.

Alphabet, digits, punctuation and symbols

a b c d e f g h i j k l m 2.1 n o p q r s t u v w x y z 2.2

A B C D E F G H I J K L M 2.3 N O P Q R S T U V W X Y Z 2.4

0 1 2 3 4 5 6 7 8 9 2.5 . , ; : ! ? ' " () [] { } 2.6

- - / \ @ # % & * + = < > € \$ § ° 2.7

Pangrams

The quick brown fox jumps over the lazy dog. 2.8

Sphinx of black quartz, judge my vow. 2.9

WALTZ, BAD NYMPH, FOR QUICK JIGS VEX. 2.10

Characters that look alike

Write each group with its characters apart, then the words below, as you would in a note.

rn m 2.11 cl d 2.12 0 0 o 2.13 l l I | 2.14 u v 2.15 a o 2.16

5 S 2.17 2 Z 2.18 g q 9 2.19

modern 2.20 clad 2.21 Oslo 2030 2.22 Illinois 2.23 vacuum 2.24

avocado 2.25 SOS 55 2.26 quiz 99 2.27 ll I l | l 2.28 00 o0 Q0 2.29

B. Running text

Copy each sentence into the box below it, on one line, in your normal size. Numbers, dates, units and addresses as printed.

The meeting starts at 9:30 in room 2.14.

3.1

Today we discuss the budget for next year.

3.2

Please read chapter 4 before the lecture.

3.3

The sensor reads 23.5 °C and 41 % humidity.

3.4

We agreed to test the new design in May.

3.5

The bridge is 340 m long and 12 m wide.

3.6

Send the slides to lena.marsh@example.org

3.7

Notes: <https://example.org/hwr/notes>

3.8

Lecture 7: signals, noise and filters

3.9

Deadline for the report: Friday, 3 June 2027.

3.10

Approx. 2.5 kg of flour, 1 l of milk, 6 eggs.

3.11

Questions? Ask Dr. Weber after class.

3.12

The error dropped from 8.4 % to 3.1 %.

3.13

B. Running text (continued)

Copy each sentence into the box below it, on one line, in your normal size. Numbers, dates, units and addresses as printed.

Call me back at +44 20 7946 0123.

4.1

Next steps: draft, review, then print.

4.2

The train leaves at 17:05 from platform 3.

4.3

Our plan for Q3: fewer meetings, more focus.

4.4

A short summary fits on one page.

4.5

Bring a pen, a ruler and your student ID.

4.6

Results vary between 0.75 and 1.25 V.

4.7

Room booked Tue 14:00-15:30, max. 12 people.

4.8

The draft goes to the editor on 2027-01-15.

4.9

Write clearly, even when you are in a hurry.

4.10

See you next week at the same time!

4.11

C. Search words

Write each word once in its box, at the size of the box: headline, large, normal and small. These words are what a search for this form looks for.

Headline

Okapi 5.1

Quoll 5.2

Ibex 5.3

Tapir 5.4

Gecko 5.5

Lemur 5.6

Large

Marrowbend 5.7

Thandeka 5.8

Ljubljana 5.9

Okonkwo 5.10

Saskatoon 5.11

Brackwater 5.12

C. Search words (continued)

Write each word once in its box, at the size of the box: headline, large, normal and small. These words are what a search for this form looks for.

QX-4471 6.1

Wexford 6.2

McAllister 6.3

isotherm 6.4

Normal

eigendecomposition 6.5

photolithography 6.6

LIDAR 6.7

OAuth 6.8

CRISPR 6.9

Gate 47B 6.10

Siobhan 6.11

Reykjavik 6.12

deGroot 6.13

hydrofoil 6.14

Route 9X 6.15

Wojciech 6.16

Small

Aotearoa 6.17

Ouagadougou 6.18

zeolite 6.19

MacGregor 6.20

NTSB 6.21

B12 6.22

quasicrystal 6.23

Ingrid 6.24

Kalmar 6.25

pH 7.4 6.26

Vexholm 6.27

counterintuitive 6.28

D. Angles

Write each line along its box, in the direction of its arrow, as if the box were level: turn the page or the tablet if that helps. The printed text is turned with its box.

→ fifteen degrees 7.1

→ tilted a little 7.2

→ slightly downhill 7.3

→ lab at 10:15 7.4

→ a gentle slope 7.5

→ the other way 7.6

→ rising notes 7.7

→ page 15 of 40 7.8

→ thirty degrees 7.9

→ a steeper note 7.10

→ room 3.07 7.11

→ thirty, going up 7.12

→ climbing words 7.13

→ step 2 of 5 7.14

→ a diagonal remark 7.15

→ halfway down 7.16

→ 45 min left 7.17

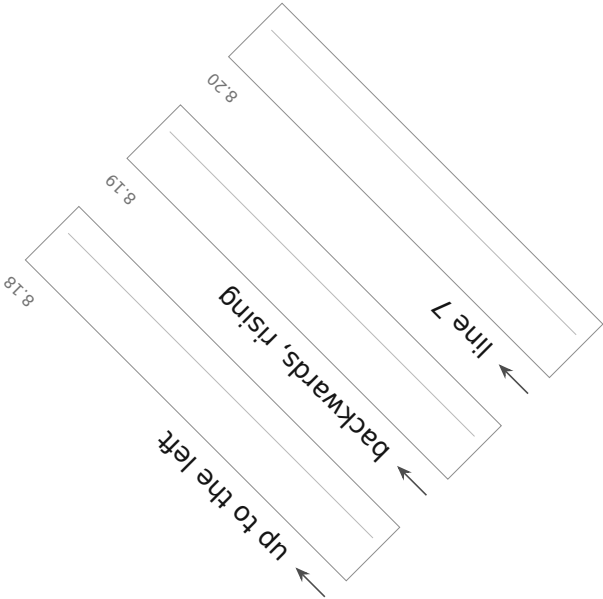
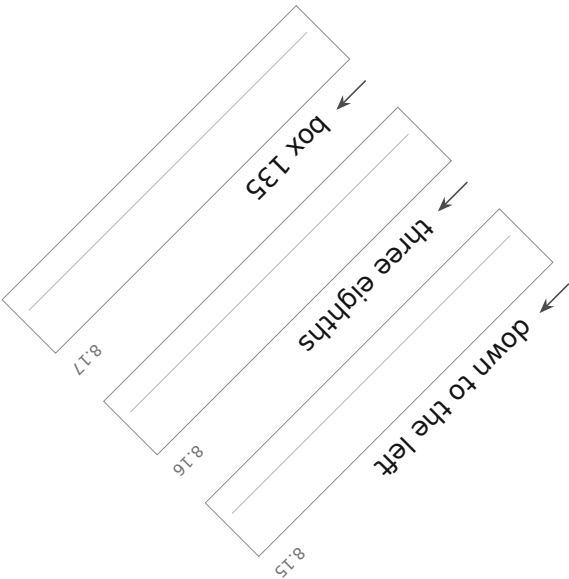
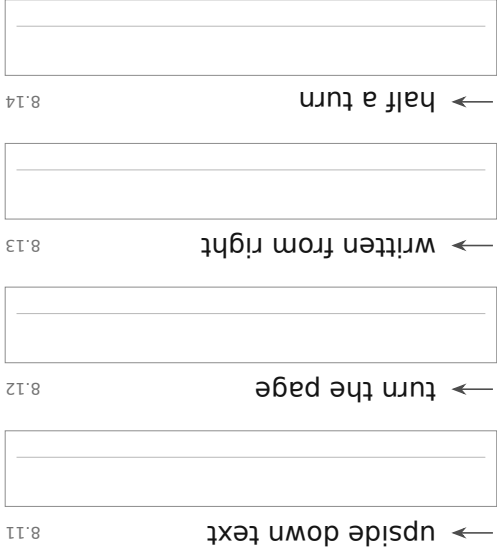
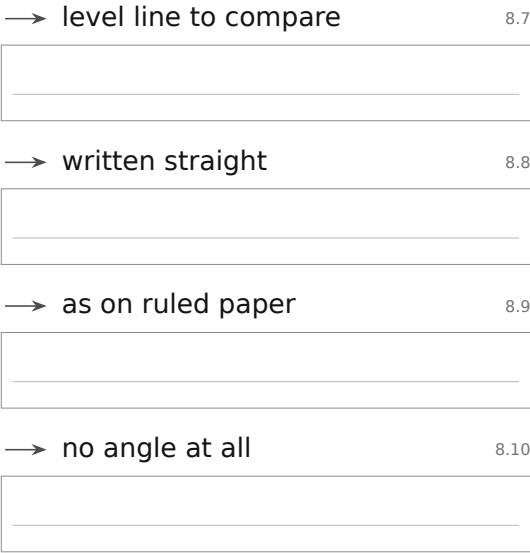
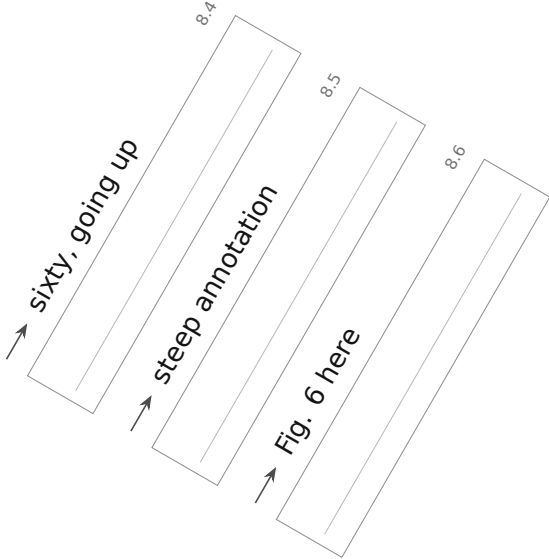
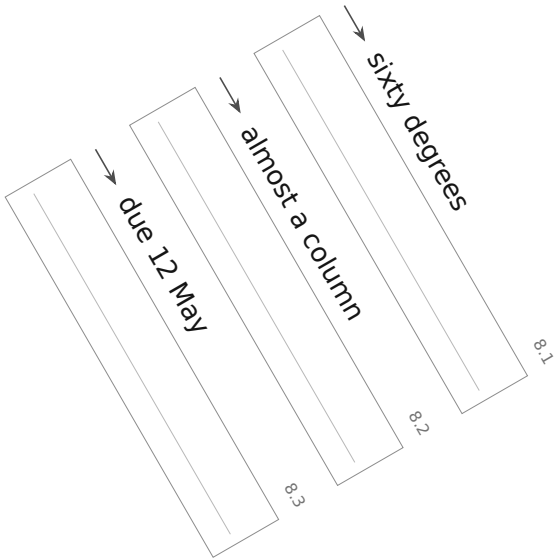
→ diagonal, going up 7.18

→ forty five 7.19

→ version 2.1 7.20

D. Angles (continued)

Write each line along its box, in the direction of its arrow, as if the box were level: turn the page or the tablet if that helps. The printed text is turned with its box.



D. Angles: short notes

Short notes of one, two and four words at every angle, the kind written beside a figure or a formula. Write each along its box, in the direction of its arrow.

→ Okay 9.1

→ see p. 12 9.2

→ Why? 9.3

→ check this again 9.4

→ True 9.5

→ not here 9.6

→ ask her on Friday 9.7

→ x2 9.8

→ too long 9.9

→ fix the second axis 9.10

→ Done 9.11

→ nice idea 9.12

→ add 3 more samples 9.13

→ Maybe 9.14

→ see below 9.15

→ this was in 2019 9.16

D. Angles: short notes (continued)

Short notes of one, two and four words at every angle, the kind written beside a figure or a formula. Write each along its box, in the direction of its arrow.

→ Wrong10.1

→ units missing10.2

→ split into two parts10.3

→ Q410.4

→ good point10.5

→ move up a level10.6

→ Later10.7

→ why so?10.8

→ compare with table 410.9

→ yes10.10

→ and then?10.11

→ same as last week10.12

→ send it to Ana10.16

→ Idea10.15

→ Upside10.13

→ room 10110.14

D. Angles: notes upwards and downwards

Short notes written downwards (arrow down) and upwards (arrow up), long notes along the page edges, and labels written upwards inside the bars. Follow each arrow.

Short notes

11.10

11.1

→ Check

11.2

→ see above

11.3

→ ask about the deadline

11.4

→ Note

11.5

→ page five

11.6

→ move this to Monday

11.11

→ And this one runs down along the right edge.

Labels inside a chart

→ This note runs up along the left edge of the page.

11.7

→ spring

11.8

→ summer peak

11.9

→ autumn

11.11

→ And this one runs down along the right edge.

E. Notes on a page

Annotate the printed text as you would a handout: margin notes (small), a word inserted between two lines, a note at the end; then a list, two columns and a table.

key idea

12.3

A filter estimates a quantity that cannot be measured directly. It combines a model of how the quantity changes with measurements that are noisy. When the model is good, the filter trusts it more; when the measurements are precise, it trusts them more.

model vs data

12.5

insert here: noisy sensor data

12.1

see lecture 5

12.4

The balance between the two is called the gain. It is computed again at every step, so the filter adapts when conditions change. The same idea guides a robot through a room and a phone through a city.

example!

12.6

Ask: how is the gain tuned?

12.2

A list

buy a new notebook for the lab

12.7

•

send the corrected figures to Priya

12.8

•

book the seminar room for June

12.9

•

Two columns

Pros: fast and cheap

12.10

Cons: heavy to carry

12.11

works offline

12.12

short battery life

12.13

easy to repair

12.14

loud fan

12.15

E. Notes on a page (continued)

Annotate the printed text as you would a handout: margin notes (small), a word inserted between two lines, a note at the end; then a list, two columns and a table.

A table: copy it into the empty grid

Sample	Mass	Date
Copper	12.4 g	3 May
Zinc	8.75 g	17 May

13.1 	13.2 	13.3
13.4 	13.5 	13.6
13.7 	13.8 	13.9

F. Drawings and marks

Draw, mark and label as asked. Labels are text; frames, arrows, circles, lines and sketches are not, so keep them out of the label boxes where you can.

Flow chart: write the labels, draw a frame around each, connect them with arrows

14.1

collect data	train model	evaluate

Marks

circle the printed word

14.5

Sandpiper

underline the printed word

14.6

Heronsgate

strike out the printed word

14.7

Bitterling

write the word in the middle, then circle it

14.8

Tarragon

Juniper

14.10

14.11

then underline it

Sketch: draw a small house

14.12

Hatch this whole area with lines

14.13

Speed over time

14.14

Plot: draw two axes and a rising curve

14.15

14.16

speed in m/s

time in s

14.17

G. Formulas

Write each formula as you would in your notes, on one line where you can. Formulas are a benchmark only; they are not used for training.

$$a^2 + b^2 = c^2$$

15.1

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

15.2

$$f(x) = 3x^2 - 2x + 1$$

15.3

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

15.4

$$\int_0^1 x^2 dx = \frac{1}{3}$$

15.5

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

15.6

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

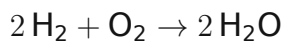
15.7

$$E = mc^2$$

15.8

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

15.9



15.10

$$R = 3,5 \text{ k}\Omega$$

15.11

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

15.12